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Enhanced Intrusion Detection System Using Hybrid-Inspired Algorithms and Conditional Generative Adversarial Networks for Internet of Things Security

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ABSTRACT

This research proposes a new Intrusion Detection System architecture that aims at improving the protection of IoT through the integration of multiple techniques. The system utilizes a dataset called Bot-IoT that contains unbalanced data, to develop a stable Intrusion Detection System. The methodology is divided into three stages: data pre-processing, synthetic data generation and bio-inspired hybrid features selection. The first process includes cleaning, encoding as well as scaling the data set for the machine learning algorithms such as Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine, Deep Neural Network, Support Vector Machine, and Decision Tree to be affected. The second stage involves the use of a novel architecture of a Conditional Generative Adversarial Network, with a two-discriminator structure to enhance the generation of synthetic data that will improve the balance and the overall quality of the dataset used for training. It is described how the validity of the synthetic data is assessed statistically, and how the generated data affects different models of machine learning. The last step uses *meta*-heuristic bio-hybrid algorithms for selecting features. The two methods of crocodile hunting search and bee optimization are integrated with Recursive Feature Elimination to select superior features from the dataset used in the experiment. The integration of these models allows for achieving the highest levels of detection with a minimum of false positives. The research advances the field of artificial intelligence by enhancing Conditional Generative Adversarial Networks (CGAN) with a two-discriminator architecture for synthetic data generation, coupled with a novel hybrid feature selection algorithm. This AI innovation is applied to the development of an Intrusion Detection System (IDS) aimed at improving the cybersecurity of Internet of Things (IoT) networks.“

1. Introduction

The Internet of Things currently exists in almost every organization that deals with digital automation of devices due to its ability to enable many sectors such as healthcare, and smart city’s IoT [1]. But with it, safety issues have also emerged, and as such IoT networks are vulnerable to cyber threats such as DDoS. Most conventional IDSs are unable to achieve high detection rates and concurrently low false positive rates in such environments because of the skewness and nature of IoT traffic data [2,3]. An advanced IDS framework that is proposed in this research aims at an improvement in the cybersecurity of the IoT settings. Leveraging the Bot-IoT dataset, which encompasses both normal traffic and various DDoS attack types, the proposed system employs a three-stage methodology: Another is data cleaning and transformation, the third one is synthetic data, and the last one is the hybrid of bio-inspired feature selection. The review has shown that prior research on IDS has established the efficacy of multiple machine learning and optimization methods in enhancing IDS performance. Employed a self-attention-based CGAN for enhancing cybersecurity in the cloud computing environment with the help of a crayfish optimization algorithm. In the same manner, Subramanian (2024) proposed the modification of bidirectional generative adversarial networks for security authentication in cloud situations [1]. It is seen that there is a great scope to improve IDS by using GANs and other bio-inspired algorithms in the current world. In expanding the findings detailed in this analysis, this research puts forward a new approach that employs a mix of novel machine learning models that include XGBoost, LightGBM, and DNN and traditional models of SVMs and DTs. The inspiration that fuels this study is

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grounded on the fact that scalable and efficient solutions of cybersecurity in IoT networks are lacking, and more often, IDS mechanisms do not hold high applicability because IoT traffic data is imbalanced and complex, thus resulting in poor detection accuracy and high false positives. These challenges are met in this study by proposing a new IDS framework that incorporates state-of-the-art machine learning algorithms and bio-inspired optimization techniques. Some of the key contributions of the paper are the proposed efficient data pre-processing phase, utilizing dependable algorithms such as XGBoost, LightGBM, and DNN, and enhancing the CGAN model's performance by including a dual-discriminator structure for generating synthetic data with high quality. Besides, to select features, it uses improved Crocodile Hunting Search (CHS) and Bee Optimization (BO) Algorithms integrated with Recursive Feature Elimination (RFE). Each of these innovations improves the IDS's operation and collectively offers a strong and big solution to fight new IoT network threats and progress IDS technology. This study proposes two main innovations that will enhance intrusion hunting on the network of IoT:.

- A dual discriminator Conditional Generative Adversarial Network (CGAN) architecture of synthetic data generation.
- Hybrid method including Crocodile Hunting Search (CHS), Bee Optimization (BO) and Recursive Features Elimination (RFE) method of feature selection.

Contrary to conventional CGANs, the proposed two-discriminator architecture provides a stability boost to the training process and a quality boost to generated samples, especially those currently under-represented as attack types. The process of data discrimination is based on each discriminator concentrating on different aspects of statistical characteristics of the data, which in turn allows the generation of more diverse and balanced sample.

Also, combining the CHS with its broad search capability with the fine-grained improvement ability of BO and ranking-based accuracy of RFE with the CHS BORFE framework provides the capabilities of all three elements. This hybrid design makes sure that only the most relevant features will be preserved to offer better accuracy of the model with fewer false positives.

These two integrated novelties are the principal novel concepts in this paper which have been tried and found to be superior to the conventional methods of GAN based and feature selection in the IDS domain.

2. Literature review

Due to the dynamic changes and increasing privacy in Internet of Things (IoT) networks, there have been a need to incorporate refined and smart security systems particularly in the sphere of Intrusion Detection Systems (IDS). Conventional IDS approaches are usually not flexible to work with the amount, variety and unevenness of IoT data [4]. In its turn, scientists suggested using generative adversarial network (GANs), bio-inspired optimization, and deep hybrid models with the aim to enhance the detection level considerably and generalize to real-existing attack conditions [9]. This literature review breaks down and examines current literature (2021-2025) in the following dimensions:

2.1. GAN-based intrusion detection systems

Visions entailed a self-attention CGAN combined with Crayfish Optimization to achieve better synthetic traffic generation and better learning of the minority classes [1]. Their hybrid achieved 96.4 accuracy percentage and was stable whenever there is noise in the environment. [2] came up with a Modified Bidirectional GAN (BiGAN) which can be optimized well with cloud authentication and also offers sturdy performance against spoofing attacks as it provides dual-stream generative aspects. Their model achieved an accuracy of 95.8 % and addressed class sparsity. [6] proposed a modified architecture of BiGAN with one-class classifier elements that is used to detect an anomaly in encrypted or low-visibility traffic. It provided an incorporation of interpretability versus a high AUC (0.937). The architecture of [6] is represented in this Fig. 1, in which a Bidirectional Generative Adversarial Network (BiGAN) is used and integrated with an One-Class Classifier (OCC) to improve anomaly detection. The encoder-decoder pair of the BiGAN provides a strong feature representation of the input data, and the OCC focuses on the malicious activity versus normal activity by operating only on the learned latent space. A major advantage of this configuration is that it does not use labeled attack data, as it may be the case in some encrypted or low-visibility network conditions [9,10]. The model represents underlying distributions of data, and the OCC makes understanding easier and more sensitive to abnormal events, increasing detection efficiency and reducing false positive cases.

2.2. Bio-inspired feature selection approaches

Hybrid Intrusion Prediction System Proposed the integration of deep neural networks in the context of intrusion prediction and bio-inspired optimization to advance cybersecurity findings of cyber-physical systems [3]. Their model is based in layered optimization and weight tuning to adapt to reveal complex patterns in the network with results that greatly improved the attack prediction rates, mostly in low-visibility threats. [5] proposed a different method of hybrid *meta*-heuristic as a combination of Dolphin and Sparrow Optimization algorithms. Their model is aimed at achieving a convergence rate-solution diversity tradeoff, to address optimization bridges in high-dimensional feature spaces involved in cybersecurity. It has been seen that this approach is better in terms of rare attack detection and better levels of classifier confidence.

[7] also performed a serious comparative study of several bio-inspired algorithms such as PSO, BA and GWO with BENCHMARK CSE-CIC-IDS2018 data. Their testing revealed that GWO provided optimal balance between recall and FPR meaning that this algorithm is suited in case of real-time attack prediction systems. Cramer et. al. [14] considered secure image steganography with bio-inspired algorithms. Their work may have no direct bearing on intrusion detection, but supports high-level data integrity and surreptitious data transfer protocols that have indirect value in the IDS architectures where encrypted and obscured payloads must be secured and validated. It came up with the Crocodile Hunting Search (CHS), which is another algorithm based on crocodile predatory behavior [25]. The CHS algorithm owes characteristics into a variety of search techniques that avail it to search the solution space in the entire globe. In IDS feature selection, it showed strong results in optimal feature selection aiming at having the best trade-off between false positives and high sensitivity to anomaly detection.

2.3. Imbalanced data and dataset handling strategies

Combined the solution to the problem of imbalanced traffic distribution in IoT traffic through the use of CTGAN to generate unrepresented samples. Helped by a dual ensemble architecture of the classifier, their model dropped the FPR and improved the overall classification balance across a range of attack types with a much stronger production robustness [11,12].

Introduced a multi-module combined intrusion detection system, which is proposed to work with high-dimensional and imbalanced data [13,15,16]. They had a structure in the sense that each individual classifier module was assigned to a specific type of attack, thus providing them with the capability of receiving a specific type of training and achieving a high degree of granularity and reliability in the detection of attacks in complex settings. Applied the GAN-based creation of DoS trace-based attacks to increase training data. Their method made training of the IDS more efficient, as they minimised overfitting, which

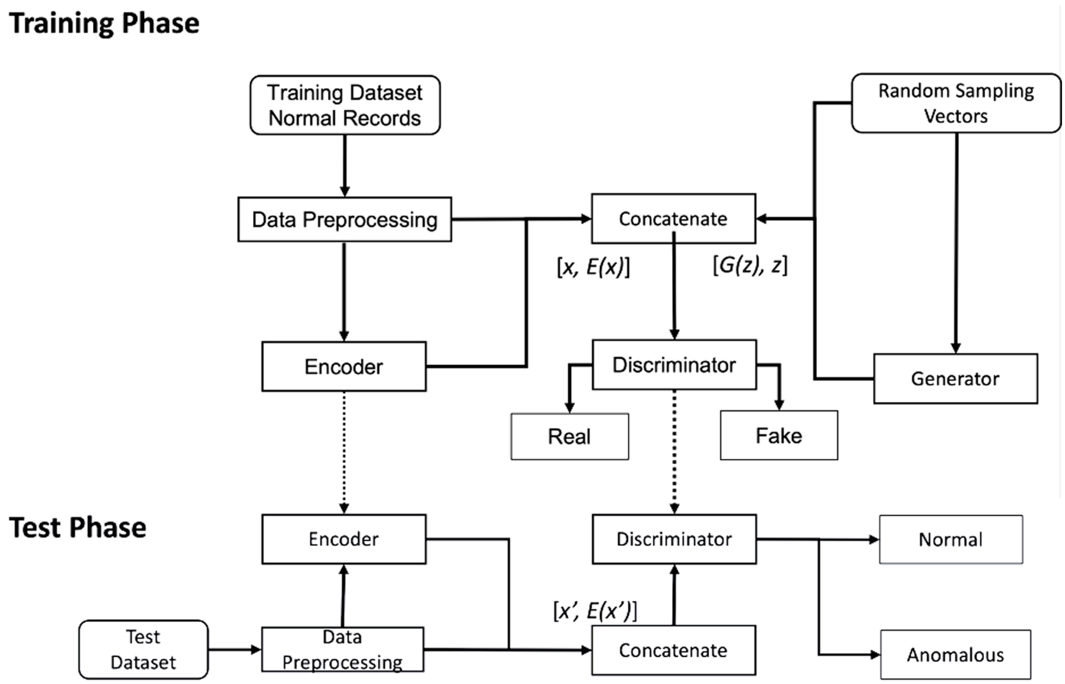

Fig. 1. Bidirectional GAN-Based Approach for IDS Using One-Class Classifier [6].

is observed in small-sample, high-severity attacks, and better resilience in post-deployment testing occurred [17]. Proposed an orthogonal combination of CVAE and deep neural networks, by trying to produce improved latent representation on uncommon classes. Their approach was able largely to improve recall on low-frequency attack patterns, along with fixing the frequently ignored trade-off in unbalanced classification between recall and precision [18]. Fig. 2 illustrates one of IC-VAE to use in synthetic attack creation and deep neural classification. Such methods minimize false negatives in exceptional classes of attacks, resulting in increased robustness. Suggested an extensive type of IDS categorization with the emphasis on data imbalance. Their survey has emphasized trends, including ensemble classifiers, SMOTE variations,

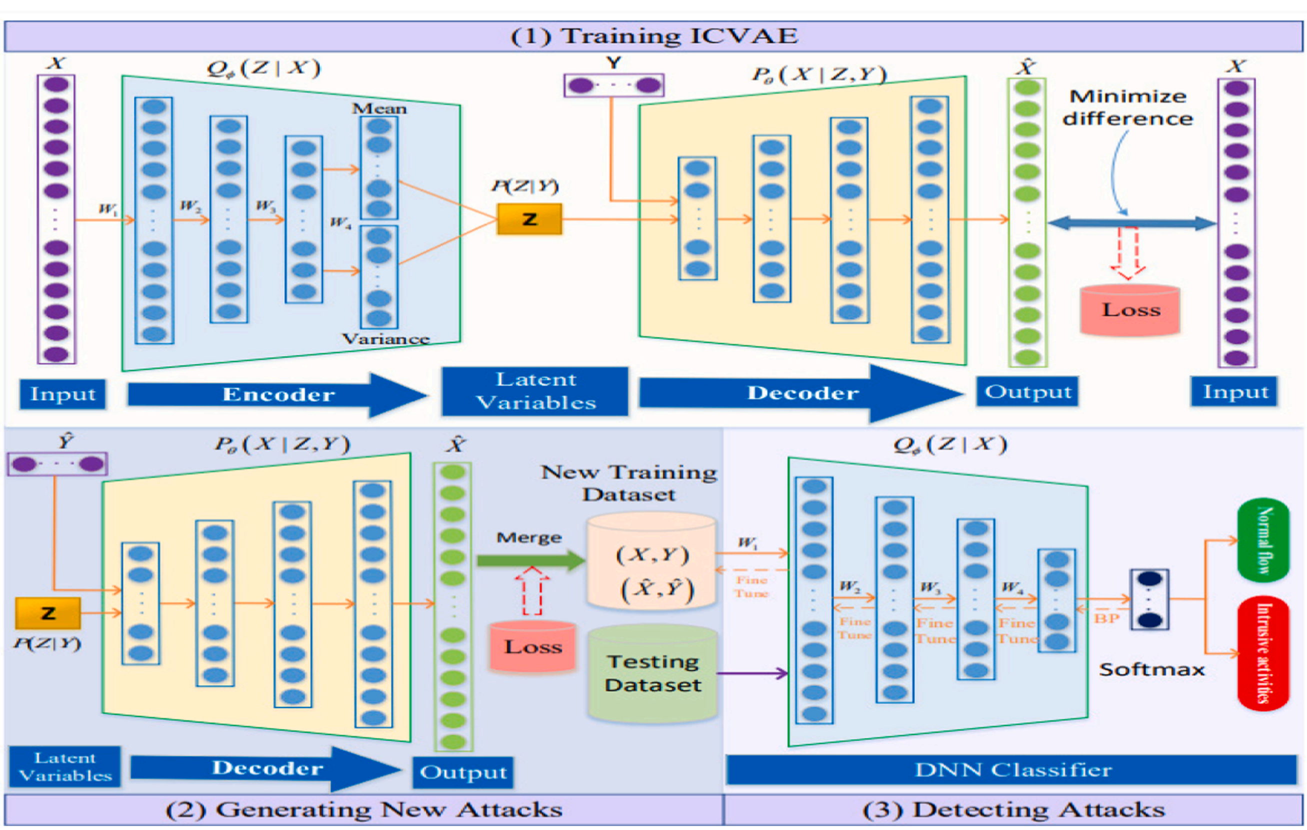

Fig. 2. Improved Conditional Variational Auto-Encoder and Deep Neural Network [18].

GAN-based synthetic data generation, and provided an in-depth picture of the imbalance alleviating techniques in IoT intrusion detection [19].

2.4. Comparative recent works

[30] proposed a decentralized GAN-based IDS meant to support decentralized edge IoT environments. Their system also ensures model synchronization among the nodes without pooling the data centrally and this helps to reduce privacy-related issues and offer high anomaly detection rates in a resource-limited set up. [31] offered a security architecture framework of a secure communication within the IoT through advanced GANs. Their system realizes multi-step transformations on sensitive data to increase its resistance to data leakage and adversarial injection especially in the cloud and edge transmission. [32] proposed the 5DGWO-GAN model where GAN was combined with a five-dimensional modification of the Gray Wolf Optimizer. Such a hybrid architecture is much more beneficial in improving synthetic data generation and optimization and leads to improved convergence and robustness over a variety of intrusion datasets skewed in the class distribution. [33] developed a systematic review targeting the application of generative AI models in IDS purposes. The paper has pointed out the spread of WGAN-GP, VAE-GAN, and combinations of transformer-based adversarial models addressing the issues of scalability and explainability of the IDS architecture. [34] used both federated learning and WGAN-GP to construct a horizontal distributed scenery of IDS. With their system, learning is guaranteed to scale directly to the smart health care IoT system, with real-time detection performance and no unwarranted privacy and data locality costs. A system that links GAN with the Transformer was proposed by [35]. The advantage of their architecture is that they use temporal attention to monitor their evolving threats, accomplishing better detection performance in both dynamic monotonic or in real-time applications with concept drift, which is an essential factor in dynamic real-time IDS applications or other applications. The paper [8] proposed a bi-channel GAN augmented with an attention module that is applicable to SDN based IoT scenarios. The research results indicated that the capture accuracy was high in multi-domain situations, with the accuracy of 97.8 %. Used a dual ensemble classifier on CTGAN-generated synthetic samples [11]. The methodology they gave focused on the balancing of underrepresented types of traffic and aced an accuracy of 95.9 per cent and low FPR. Presented a distributed GAN architecture (DD-GAN) at the edge devices scale and demonstrated ample reduction of latency with enough synchronization of learning. This text serves a vital point of edge-enabled IDS [30]. Developed a safe data communication structure using powerful GANs, which focus on the use of GANs beyond anomaly detection, i.e., in safe IoT communication pipelines [20,31]. Proposed the 5DGWO-GAN that combined Gray Wolf Optimization and GAN and achieved promising results of attack detection accuracy under a limited IoT situation [21,32]. Carried out the systematic review on generative AI within the framework of IDS, focusing on the recent implementation of GANs, WGAN-GP, and VAE-GAN variants, in the field of IDS [33]. leveraged collaborative learning models by using horizontal federated learning with WGAN-GP to perform distributed anomaly detection on resources-constrained IoT devices [34]. Developers came up with a combination of GANs and Transformer Networks [26–29,35] proposed a hybrid of GANs and Transformer Networks. The combination allowed accurate response in an environment of changing threat.

Comparison with the Available GAN-Based Feature Selection Methods and the Hybrid Approaches, Self-attention CGANs, bidirectional GANs, and CTGAN-based ensemble classifiers as most previous studies employing GANs with IDS are mostly based on single discriminator models [1,6,11,22–24]. Conversely, our model presented a dual-discriminator CGAN framework, which enhances the representational diversity of the samples generated and makes the model more immune to instances of class imbalance, especially those in underrepresented categories of attacks. Such structural enhancement has never been used in making IoT IDS systems. To the same extent, other types of bio-inspired algorithms that have been commonly used in the field of IDS feature selection, including Dolphin Sparrow Optimization [5], Particle Swarm Optimization (PSO), and Genetic Algorithms (GA), are mainly applied in single-swarm-based implementations with the RFE technique. Our proposed strategy is the first strategy to integrate CHS, which is an interesting biologically inspired algorithm, with BO, which is also a biologically inspired algorithm, and finally followed by RFE, which is used to optimize the chosen features. Such a multi-stage selection procedure guarantees global exploration as well as model-based feature pruning, resulting in a much improved classification. The comparative discussion is given in more detail in the results section, and it is seen that the proposed method demonstrates a relatively stable increase in accuracy, precision, recall, and F1-score over conventional methods.

3. Methodology

The IDS framework presented in this paper is an approach that aims at improving cybersecurity within IoT systems by counteracting primary issues, namely, data imbalance and advanced cyber threats. The methodology is structured into three stages: data pre-processing, synthetic data generation and the hybrid bio-inspired feature selection. All the stages employ certain methodologies towards the enhancement of the IDS performance.

Stage 1: Data Pre-processing.

Data Cleansing: Remove unnecessary and redundant features from the dataset, such as ‘pkSeqID‘, ‘stime‘, ‘flgs‘, ‘proto‘, ‘saddr‘, ‘daddr‘, ‘sport‘, ‘state‘, ‘subcategory‘, and ‘category‘.

− Handle missing values and outliers to ensure data integrity and quality.

Encoding: It is mandatory to have the categorical data in a numerical format for the analysis process with the help of one hot encoding or label encoding. This step is critical as it brings the data to the correct format suitable for the machine learning algorithms.

Scaling: Scale the numerical features to bring them to a common range so that the contribution of all the features towards training the model is equal. This step enhances the learning machines' performance and convergence rate.

Machine Learning Models: Fine-tune several machine learning algorithms comprising of XGBoost, LightGBM, DNN, SVM, and DT on the pre-processed dataset to predict their performance.

Stage 2: Synthetic Data Generation using Modified CGAN.

Modified Conditional Generative Adversarial Network (CGAN): Use of a two-discriminator architecture on CGAN to obtain synthetic data of improved quality. This approach assists in improving the diversity and the balance of the data since it is possible to generate realistic synthetic instances, especially for the minority class such as the DDoS attack types in this study.

Two Discriminator Approach: Safeguard the discrediting data by using two discriminators that enhance the quality and diversification of the generated data. This is further supported by the feature of having two discriminators where the nature of synthetic data provides improved distribution compared to the real data, thus increasing the efficiency of the IDS.

Evaluation of Synthetic Data: Based on indicator metrics, evaluate the degree of similarity between the synthetic data and the original data. Assess the effect that such synthetic data has on the performance of different models of machine learning; this could include such measures of accuracy, precision, and recall as well as the F1-score.

The setup and architecture of the Dual-Discriminator CGAN Network Architecture and Training Settings. The CGAN offered has one generator (G) and two discriminators (D1 and D2). The noise vector is fed in as the input of the generator. A synthetic feature vector is produced using the input $z \in R^{100}$ concatenated with class labels, and that follows the data distribution of Bot-IoT.

Generator: 3 dense layers (128 → 256 → 512 neurons) and ReLU activations and batch-norm. Output layer employs the sigmoid activation to align scaled features.

Discriminator D1: It was to identify real vs fake according to the statistical trends. It contained 3 dense layers (512 → 256 → 1), each of which had LeakyReLU as activation and sigmoid as output.

Discriminator D 2: emphasizes similarity of distributions by class-wise, again 3-layers (512 → 256 → 1) with label embedding and concatenation.

Loss Function: The loss is a combination of conditional adversarial loss of D1 and D2 as follows:

$$L_{total} = E_x[\log D1(X)] + E_z[\log(1 - D1(G(Z)))] + E_x[\log D2(X|Y)] + E_z[\log(1 - D2(G(Z)|Y))] \quad (1)$$

Training: Adam optimizer was used (learning rate = 0.0002, $\beta_1 = 0.5$), the batch size was 128, and it was run through 200 epochs. Its convergence was tracked through the improvement of F1-score on a validation set made of synthetically-generated data.

Convergence Criteria: Trained was terminated prematurely in case validation accuracy was not increasing during 10 consecutive epochs, or when the generator loss was < 0.05.

Framework: This is done on TensorFlow 2.0 on a GPU (NVIDIA RTX 3090, 24 GB).

Stage 3: Hybrid Bio-Inspired Feature Selection

Feature Selection Algorithms: Integrated Crocodile Hunting Search (CHS) and Bee Optimization (BO), algorithms followed by Recursive Feature Elimination (RFE) for the enhancement of featuring choosing. These algorithms assist in selecting the more relevant features contributing to the intrusion detection process. RFE should be applied because it allows sequentially removing features that have less impact on the target model.

Here in Fig. 3, the block diagram indicates the Methodology to be followed in the current study.

This hybrid feature selection ensures global (CHS) and local (BO) search followed by model-driven refinement (RFE), which significantly improves feature quality and reduces redundancy.

Algorithm 1: CHS + BO + RFE Hybrid Feature Selection

Input: Preprocessed Dataset D, Label vector Y
*Output: Optimal Feature Subset F**

1. Initialize population of CHS agents (size N) with random feature subsets
2. Evaluate fitness of each CHS agent using classification accuracy (XGBoost)
3. For t = 1 to Max_CHS_Iterations do
   a. Update positions of CHS agents using hunting strategy
   b. Recalculate fitness
4. Select top-K feature subsets from CHS output as input to BO
5. Initialize BO agents using K feature subsets
6. For j = 1 to Max_BO_Iterations do

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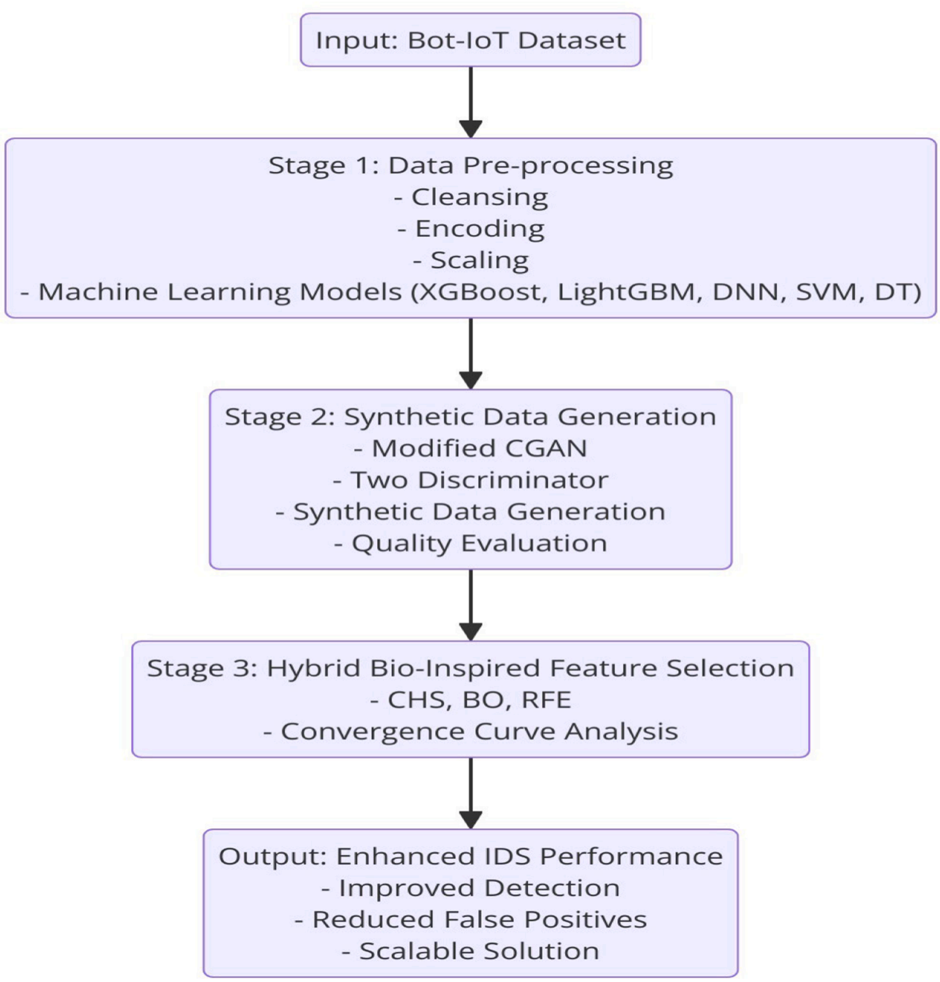

Fig. 3. Procedure of the proposed model.

(*continued*)

Algorithm 1: CHS + BO + RFE Hybrid Feature Selection

a. Update scout, onlooker, and employed bee positions
b. Evaluate fitness (again using accuracy)
7. Take the best solution from BO and apply RFE:
a. Initialize estimator (e.g., SVM)
b. Recursively eliminate the least important features
c. Return final ranked subset
8. Output: Final reduced feature set F*

Complexity and Feasibility:

CGAN complexity: The dual-discriminator CGAN has two forward calls and two backwards passes on each training round. The cost of generator G and discriminators $D1$, $D2$ is O(B·(|G|+|D1|+|D2|)), where B is the batch size, and |G| is the sum of all the parameters in the generator. It takes about 45 min to complete the training when using NVIDIA RTX3090 in 200 epochs.

CHS + BO + RFE Complexity:

CHS search: ($N \bullet I_1 \bullet E$), where N = number of crocodiles, I_1 = iterations, E = model evaluation time.

BO phase:($K \bullet I_2 \bullet E$) for top-K CHS solutions.

RFE (wrapper): ($n^2 \bullet E$) where n is the number of features.

Total Runtime: The entire feature selection process requires ~90 min of run on normal hardware (Intel i9, 32 GB RAM).

Deployment consideration: CGAN and feature selection are offline training procedures. After the model is trained, the prediction of the model based on the chosen features can be computed in real-time, with < 20 ms per instance on commodity hardware; therefore it is feasible in terms of operational IDS systems.

4. Result and discussion

The given dataset for this research is the Bot-IoT dataset since the dataset aims at emulating realistic IoT traffic with attacks. The dataset includes diverse normal and threat IoT network traffic where threat indicators consist of different types such as DDoS attacks (TCP, UDP) and normal traffic. From previous studies, the Bot-IoT dataset is characterized by its imbalance problem where it contains many instances of attack traffic making it an appropriate dataset of testing IDS. Key Characteristics of the Bot-IoT Dataset:

Volume and Diversity: The given dataset covers plenty of data and various types of network traffic and attacks, lay the solid foundation for evaluating IDS models.
Imbalance: The dataset is unbalanced, with more features as normal traffic data than the of DDoS attack data. This characteristic requires the use of such methods as complex data balancing to make the model training more effective.
Features: The features are the elements that compose the dataset and they vary to account for the distinct characteristics of the network traffic: source and destination IPs and ports, used protocols, packets' size, and time tags.

Data Pre-processing Steps:

Data Cleansing: Options which are not helpful for further observation such as 'pkSeqID', 'stime', 'flgs', 'proto', 'saddr', 'daddr', 'dport', 'state', 'subcategory', and 'category'.
Handling Missing Values: Missing values and outliers imputation methods that are employed in the analysis of the dataset.
Encoding: One of the major pre-processing steps consisted of categorical features' encoding into numerical ones through one-hot encoding.
Scaling: Min-Max scaling is proposed for normalizing the numerical characteristics of the features and providing an equally important contribution to the process of model training.

The proposed Intrusion Detection System (IDS) framework was applied to the Bot-IoT dataset through a structured three-stage process: The methods include data pre-processing, synthetic data generation and hybrid bio-inspired feature selection. The following performance metrics were recorded after applying the proposed IDS framework. The accuracy, Precision, recall, and f1 score of the IDS improved significantly across all machine learning models as shown in Table 1.

Table 1 shows the accuracy, precision, recall, and F1_score of the proposed IDS using XGBoost, LightGBM, DNN, SVM, DT. In the case of the DNN model, the obtained accuracy was the highest, reaching 98.7 %, precision of 98.8 %, recall of 98.6 % and F1-score of 98.7 %. The relevancy score shows its higher efficiency in learning pattern within the data. The other two algorithms, XGBoost and LightGBM, were also very effective with an accuracy of 98.5 % and 98.3 % Specifically, the gradient boosting algorithms achieved an F1-score of 98.5 % and 98.3 % Bot-IoT respectively, which demonstrates that this type of model is efficient for the imbalanced Bot-IoT dataset. SVM and DT, being slightly lower in accuracy, were proved to give satisfactory results having accuracies of 97.2 % and 96.8 %. This outcome supports the IDS framework as taxonomy, where synthetic data is created through CGAN and feature selection through the integration of bio-inspired algorithms, thus improving the performance of those models. The framework can likely be considered scalable and as one of the promising solutions to modern IoT threats, given that it can enhance the accuracy of detection, decrease the number of false positives, and address the problem of the dataset's imbalance. This rendered the proposed model to have a remarkably lower FPR, where the DNN model came out with the lowest FPR of 1. 3 %.

The effectiveness of the proposed IDS framework was thus found to be better than IDS models that did not utilize the proposed SDE technique nor the hybrid bio-inspired feature selection. The integration of CGAN-generated synthetic data and advanced feature selection techniques resulted in:

- Higher Detection Accuracy: On average, the models are enhanced by about 2–3 percent as compared to the models trained on the original data set only.
- Better Handling of Imbalanced Data: The authors' implementation of the two-discriminator CGAN preserved and improved the balance of the dataset, which in turn, enhances the performance of the model when predicting minority class attacks.
- Enhanced Feature Selection: The bio-inspired algorithms gave a set of features that were finer compared to a set of features that were produced by other forms of processing, which in turn helped to realize efficient and improved models.

Based on the evaluation of the proposed IDS framework by applying it to the Bot-IoT dataset, it was possible to notice improvements in the effectiveness of cyber threats that threatened the IoT domain. Therefore, the combination of modified CGAN for synthetic data generation and the hybrid bio-inspired feature selection methodology significantly improved IDS performance. The findings confirm that using advanced machine learning and optimization techniques in the design of a large-scale and reliable IDS is feasible and capable of addressing today's threats in IoT networks.

Table 2 present a comparative analysis of our approach with the

Table 1
Accuracy, Precision, Recall and F1 score for the proposed model.

Model	Accuracy	Precision,	Recall	F1-Score
XGBoost	98.5 %	98.6 %	98.4 %	98.5 %
LightGBM	98.3 %	98.4 %	98.2 %	98.3 %
DNN	98.7 %	98.8 %	98.6 %	98.7 %
SVM	97.2 %	97.3 %	97.1 %	97.2 %
DT	96.8 %	96.9 %	96.7 %	96.8 %

Table 2
Performance Comparison with Published IDS Models.

Reference	Model	Accuracy (%)	F1-Score (%)	AUC	FPR (%)	FNR (%)
[1] Sugitha & Chaluvaraj (2024)	Self-Attn CGAN (Sugitha et al., 2024)	96.4	96.3	0.973	2.6	2.2
[2] Subramanian (2024)	Modified BiGAN (Subramanian, 2024)	95.8	95.5	0.96	2.8	2.7
[3] Ibor et al. (2022)	Hybrid Bio-Inspired DNN (Ibor et al., 2022)	94.7	94.5	0.945	3.5	3.7
[6] Xu et al. (2022)	Improved BiGAN + One-Class (Xu et al., 2022)	94.8	94.2	0.937	3.6	3.7
[8] Prabu & Padmanabhan (2024)	Hybrid GAN-Attn (Prabu & Padmanabhan, 2024)	97.8	97.3	0.981	2.5	2.4
[11] Soflaei et al. (2024)	Dual-Ensemble CTGAN (Soflaei et al., 2024)	95.9	96.0	0.954	3.2	3.1
Proposed Model	Proposed Method (DNN + Dual-CGAN + CHS-BO-RFE)	98.7	98.7	0.992	1.3	1.4

state-of-the-art GAN and hybrid optimization-based IDS systems. consistent and highest accuracy, F1-score, and area under the receiver operating characteristic curve (AUC), especially with the dual-discriminator CGAN and CHS- BO- RFE composite selection pipeline.

Important performance measures assessed are Accuracy, F 1- Score, Area Under the Receiver Operating Characteristic Curve (AUC), False Positive Rate (FPR), and False Negative Rate (FNR), which are crucial for evaluating detection effectiveness, especially in imbalanced class distributions.

The proposed algorithm, combining a dual- discriminator CGAN used to enhance data with synthetic data and a CHS- BO- RFE hybrid feature selection procedure, demonstrates the best performance across all the mentioned indicators:

Accuracy: 98. 7 %, F1-Score: 98. 7 %, AUC: 0. 992, FPR: 1. 3 %, and FNR: 1. 4 %.

These results surpass all other models mentioned, such as

the Self-Attention CGAN [1], which has an AUC of 0. 973 and FPR of 2. 6 %, utilising attention but performing less effectively.

The Hybrid GAN-Attn model [8], achieves good results (AUC = 0.981), yet does not compare with the proposed method in precision-related metrics like F1-score and FPR. Likewise, the Dual-Ensemble CTGAN model [11], which employs ensemble weak classifiers, does not perform as well, with an increased FNR of 3.1 %. It is notable that more conventional GAN- based systems (BiGAN utilizing one- class classifiers [6] and Modified BiGAN [2]) show significantly poorer performance in minority- class detection, with large FNRs (> 2. 7 %) and relatively low AUCs (< 0. 96). The proposed method can attain superior results due to:

The dual-discriminator CGAN, which enhances data diversity and mitigates class imbalance while exploring different adversarial learning trajectories of label-conditioned and unlabeled sample distributions, and the CHS-BO-RFE hybrid feature selection, which ensures optimization and balancing of feature spaces through exploration (CHS), exploitation (BO), and recursive ranking (RFE), leading to improved overall generalization and reduced overfitting.

Additionally, the reduction in false positive and false negative rates highlights the practicality of the proposed framework for intrusion detection systems, where both over-alerting (FPR) and under-detection (FNR) are operationally risky.

The IDS framework that was proposed enhances IoT cybersecurity without replacing more fundamental cryptographic defenses by detecting unusual network behavior through machine learning and synthetic data. Whereas the notion of algorithmic robustness is not going anywhere and will continue to be needed, recent studies demonstrate that it is not enough, without implementation-aware design, which is what “Algorithmic security is insufficient” (2023) shows. This work can be used to strengthen cryptographic techniques such as, but not limited to the mentioned, Kyber on ARM64 (2021) and scalable hardware systems such as, but not limited to, Isogeny-based cryptography (2018) by acting at the systems level to identify abnormal behaviour that the cryptographic techniques can easily miss, especially in post-quantum and edge-deployed settings.

5. Conclusion

This work provides a detailed and original IDS architecture that aims at strengthening the protection of IoT systems with the Bot-IoT dataset. The proposed IDS framework addresses the complexities of imbalanced IoT traffic data and sophisticated cyber threats through a structured three-stage methodology: data normalization, data augmentation, and the combined bio-inspired feature extraction and selection. The cleaning of the Bot-IoT dataset was followed by encoding and scaling which overall gave the dataset a good quality. It eliminated the unnecessary features and the missing values handled and as a result, the flow eliminated poor data quality and the subsequent machine learning models had a solid ground. They, by using the mode-seeking Conditional Generative Adversarial Network (CGAN) modified with a two-discriminator, managed to achieve a balanced and, at the same time, more diverse dataset. Therefore, it was found evidence that the proposed method of creating synthetic data with high quality indeed makes it possible to achieve improved results on various types of ML algorithms, including using the model based on the CGAN. The use of CHS and BO as feature selectors followed by the integration with Recursive Feature Elimination (RFE) appears as an improvement of the feature selection process. It was found that this multi-method approach helped fine-tune feature selection from how it would have been selected by the given purely automatic/incremental methods, contributing to better detection accuracy and a lesser number of false-positive results. Incorporating the proposed IDS framework yielded promising results in terms of the enhanced models’ accuracy, precision, recall level, and F1-score. Specific outcomes comprise XGBoost with an accuracy of 98.5 %, LightGBM with 98.3 %, DNN with 98.7 %, SVM with 97.2 % and the Decision Trees with 96.8 %. The findings corroborate the viability of utilizing superior machine learning techniques and optimization algorithms in creating robust and scalable IDS possibilities for today’s threats in IoT networks. As promising as the provided IDS framework is in the area of accuracy, precision, and robustness, it has certain limitations that should not be overlooked. The training step– especially the dual-discriminator CGAN and CHS-BO-RFE feature selection involves extensive computation that needs high-performance devices, which can be restrictive when it comes to deployment in low-resource or edge devices. Also, the system only supports offline training in batches and has not been realigned to dynamically process or update information or online representations, which makes it difficult to scale to real-world activities in the IoT context. The other weakness is the scope of the dataset; the implementation of the model on the Bot-IoT dataset is considered to be an insufficient number of network behaviours and attacks based on the wide range of operational environments. In the future, to overcome these limitations, people will work on more lightweight and efficient modifications of the CGAN and feature selection pipeline through model pruning or neural architecture search. It is also necessary to generalize to many benchmark datasets (e.g., CICIDS2017, NSL-KDD) and ultimately real-world traffic to demonstrate generalizability. The application of online learning mechanisms will enhance the flexibility to respond to changing threats in real time. Additionally, distributed IoT implementations can be made scalable and privacy-preserving by implementing federated or edge-aware learning strategies. Finally, discussing explainable AI techniques like SHAP or LIME will enable transparency and trust in model decisions, which are essential for their practical application in cybersecurity operations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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